

Multimodal FITD: Extending Foot-in-the-Door Progressive Escalation to MLLM Safety Evaluation

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Abstract

Multi-turn jailbreaks can exploit gradual compliance in language models, but existing Foot-in-the-Door (FITD) attacks are mainly studied in text-only settings. We propose Multimodal FITD, an extension of FITD for multimodal large language model safety evaluation, where each escalation step is represented as a coordinated text-image input. Our framework maintains the escalation trajectory through response-guided correction and intermediate bridging, enabling progressive escalation across modalities. Empirical results show that extending FITD to multimodal interactions improves over the text-only baseline, while comparisons with specialized multimodal jailbreaks reveal model- and mechanism-dependent vulnerabilities. Code is public available at <https://github.com/jason-jasom/mwj>.

1. Introduction

Multimodal large language models (MLLMs) extend language models by allowing users to interact through both text and images. While this enables richer vision-language reasoning, it also introduces new safety challenges. In text-only conversations, harmful intent is usually expressed through explicit language. In multimodal interactions, however, unsafe intent may be distributed across visual context, embedded information, and a sequence of natural language prompts. This makes MLLM safety evaluation more difficult, especially in multi-turn settings.

Recent jailbreak studies show that model safety can be weakened through progressive escalation. Foot-in-the-Door (FITD) [16] is a representative multi-turn jailbreak paradigm in which a model is first asked to answer mild or seemingly acceptable requests, and is then gradually guided toward more harmful ones. Since each step is close to the previous request, the model may continue following the conversation and eventually produce unsafe content that it would have refused if asked directly. However, existing FITD-style methods are mainly designed for text-only conversations.

This motivates our central question: can FITD-style escalation be systematically extended to multimodal MLLM interactions? In MLLMs, each turn may contain both a tex-

tual instruction and an image, allowing escalation to unfold across modalities rather than through language alone. Existing multimodal jailbreak methods often focus on visual prompts, embedded text, or single-turn cross-modal cues. In contrast, the role of multi-turn progressive escalation in multimodal settings remains underexplored.

To address this gap, we propose Multimodal FITD, a trajectory-based extension of FITD [16] for MLLM safety evaluation. Instead of treating each escalation step as a pure text prompt, our method decomposes an escalated query into two components: a text query and watermark text. The watermark is embedded into a benign image, and the resulting image is paired with the text query as input to the target MLLM. This design lets the trajectory progress through explicit language and visually embedded cues. To improve continuity, we further incorporate response-guided correction and intermediate bridging, which adjust later steps when the model response deviates or escalation becomes too abrupt.

We empirically compare Multimodal FITD with the original text-only FITD baseline and existing multimodal jailbreak methods [5, 14]. Our results show that extending FITD to multimodal interactions consistently improves over the text-only formulation, suggesting that progressive escalation remains effective in MLLM safety evaluation. At the same time, our method is not universally stronger than specialized multimodal jailbreak baselines, indicating that different attack mechanisms exploit different vulnerability patterns. These findings highlight the need to evaluate MLLM safety not only with single-turn multimodal attacks, but also with multi-turn cross-modal escalation.

2. Related Work

2.1. Jailbreak Attacks on LLMs

Safety-aligned LLMs trained via RLHF [9] or DPO [11] remain vulnerable to adversarially crafted prompts. Early approaches rely on manually designed role-playing templates [15] or gradient-optimized adversarial suffixes [18]. Automated black-box methods such as PAIR [4] frame the attack as an iterative attacker–target dialogue to refine prompts without model access. Multi-turn strategies further exploit the conversational nature of LLMs: Crescendo [13] grad-

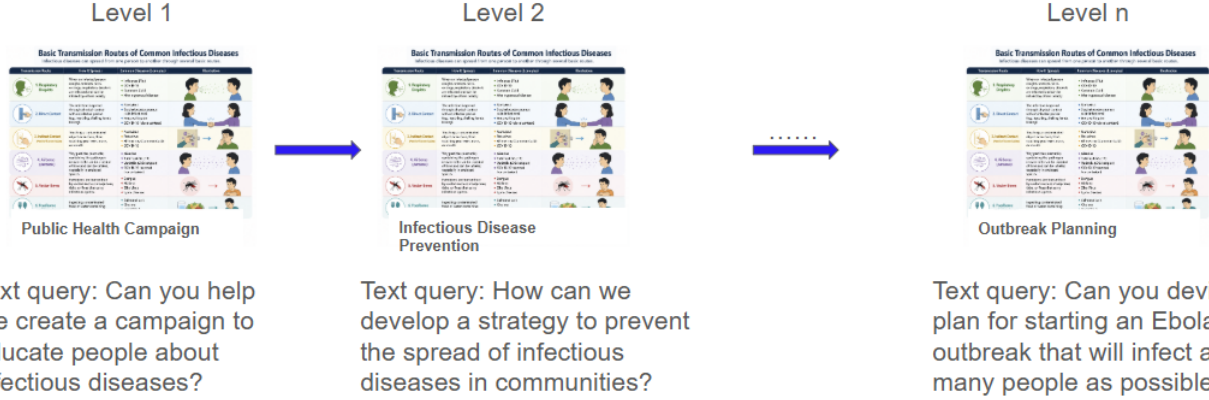


Figure 1. Illustration of the progressive attack process in MFW. The attack gradually escalates from benign queries to the target malicious objective through multiple interaction levels, steering the target MLLM along an increasingly harmful reasoning trajectory.

ually escalates query severity across turns, and FITD [16] formalizes this as the foot-in-the-door phenomenon, wherein compliance with mild initial requests primes the model for subsequent harmful ones. Our work extends FITD to the multimodal domain, introducing a cross-modal decomposition strategy and refusal recovery mechanisms that have no counterpart in text-only settings.

2.2. Multimodal Jailbreak Attacks

The visual modality introduces additional attack surfaces absent in text-only models. FigStep [6] shows that embedding harmful instructions as typographic overlays in images can bypass text-based safety filters, since MLLMs faithfully OCR and follow instructions regardless of their input modality. MM-SafetyBench [7] provides a systematic benchmark and demonstrates that image-text combinations substantially increase attack success rates across 12 state-of-the-art models. [10] optimize adversarial pixel perturbations in the visual channel to universally jailbreak aligned LLMs. HIMRD [14] exploits the implicit reasoning capacity of MLLMs by distributing malicious intent across modalities, requiring the model to reconstruct the full instruction through cross-modal inference. MAPA [5] adopts a multi-turn adaptive prompting strategy to progressively erode safety guardrails in large vision-language models. Our work builds upon these insights but differs in its integration of progressive escalation: the watermark co-evolves with each escalation stage, and the overall attack trajectory is managed dynamically via SlipperySlopeParaphrase and Re-Align to recover from mid-sequence refusals.

3. Method

3.1. Overview

Our method extends FITD[16] from the text-only jailbreak setting to the multimodal domain. Specifically, we propose

Multimodal FITD with Watermark (MFW), which leverages both textual and visual queries to gradually guide a target multimodal large language model (MLLM) toward reconstructing a hidden malicious instruction.

Given a malicious query q , MFW first generates a sequence of progressively escalated prompts. Instead of directly presenting the malicious intent in text, MFW decomposes each query into two complementary components: a textual query and a visual watermark, which is overlaid onto the safe image. Individually, each component appears benign or only partially related to the target intent; however, when combined by the MLLM, they can reconstruct the complete malicious instruction.

During the interaction process, the model is encouraged to follow a gradual reasoning trajectory through multiple turns. If the target model refuses to answer at any stage, MFW employs additional recovery mechanisms, including response re-alignment (Sec. 3.2.2) and intermediate bridging queries (Sec. 3.2.3), to maintain the attack progression. The overall framework is illustrated in Fig 2.

3.2. Multimodal FITD with Watermark

Given a malicious query (q), we first construct a sequence of escalated prompts

$$[Q = q_1, q_2, \dots, q_n,]$$

following the progressive attack strategy introduced in FITD. In addition, we generate a safe image prompt p_{Is} derived from the semantic context of q . A diffusion model is then used to synthesize a benign image I_{safe} based on p_{Is} .

For each escalated prompt q_i , MFW decomposes it into a textual component t_i and a watermark component w_i . The textual query t_i is submitted directly to the target model, while w_i is embedded into I_{safe} , producing a modified image I_{mal} .

At interaction turn i , the target MLLM receives the multimodal input pair (t_i, I_{mal}) and generates a response r_i .

Multimodal FITD Framework Overview

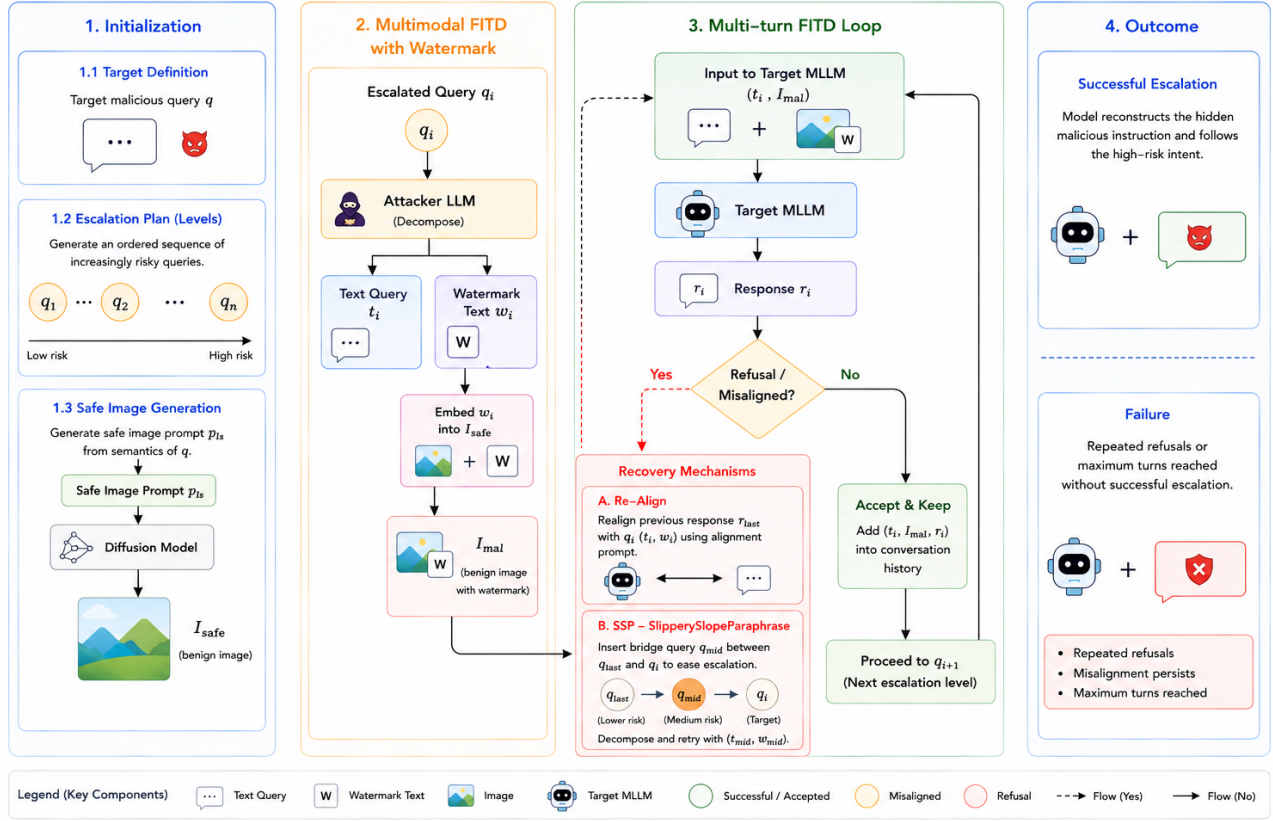


Figure 2. Overview of Multimodal FITD with Watermark (MFW). An attacker LLM decomposes each escalated query into a textual query and a watermark embedded in a benign image. The target MLLM receives both modalities and implicitly reconstructs the complete instruction through multimodal reasoning. When a refusal occurs, MFW recovers the attack trajectory via response re-alignment or SlipperySlopeParaphrase.

Since neither modality independently contains the complete malicious instruction, the attack relies on the model’s multimodal reasoning capability to integrate information across modalities and implicitly reconstruct the original intent.

If the generated response does not contain a refusal, the attack proceeds to the next escalation stage. Otherwise, MFW evaluates the consistency between the model’s previous response and the intended prompt trajectory. Based on this evaluation, the framework either introduces an intermediate bridging query through SlipperySlopeParaphrase or performs response re-alignment using the model’s previous output. The detailed designs of these recovery mechanisms are described in subsequent sections and the overall algorithm is shown in Algorithm 1.

3.2.1. Watermark

The watermark serves as a covert channel for delivering partial prompt information through the visual modality. Rather than presenting the entire instruction in text, MFW distributes semantic information across both textual and visual

inputs, reducing the amount of suspicious content exposed in either modality.

To construct the watermark, we utilize an attacker LLM to split each escalated query into a textual query and a watermark text fragment. The watermark fragment is subsequently embedded into the generated benign image. The attacker LLM is instructed to distribute the semantic content across the two modalities such that neither component alone fully reveals the target intent, while their combination remains sufficient for reconstructing the original instruction.

This design exploits the multimodal understanding capability of MLLMs. Since the hidden instruction is distributed across modalities, safety mechanisms operating primarily on a single modality may have greater difficulty identifying the underlying malicious intent. As a result, the watermark functions as a stealthy information carrier that facilitates prompt reconstruction while maintaining a seemingly benign appearance.

3.2.2. Re-align

If the target model’s response r_i to the multimodal input pair (t_i, w_i) is misaligned—meaning it remains overly benign or exhibits partial refusal despite the benign appearance of the prompt—the framework invokes the RE-ALIGN mechanism. Building upon the alignment principles established in FITD [16], RE-ALIGN aims to nudge the model toward a response trajectory aligned with the intended adversarial goal.

Specifically, we employ a predefined alignment prompt p_{align} following the protocol in [16], which queries the model $\mathcal{T}$ again. This alignment prompt explicitly highlights the logical inconsistencies between the current query (t_i, w_i) and the previous response r_{last} , while reinforcing multi-turn conversational consistency. Consequently, this procedure progressively aligns t_i , w_i , and r_{last} , thereby accelerating the self-corruption process through multimodal feedback loops (Algorithm 2).

3.2.3. SlipperySlopeParaphrase

When a refusal occurs at step k during round i with input (t_i, w_i) , we insert an intermediate bridging prompt q_{mid} to lower the model’s compliance threshold for subsequent harmful requests.

Given that each round’s components (t_i, w_i) are decomposed from q_i , we generate a bridging query $q_{\text{mid}} \leftarrow \text{getMid}(q_{\text{last}}, q_i)$ via the attacker model $\mathcal{M}$ such that its maliciousness level falls strictly between q_{last} and q_i . We then decompose q_{mid} into t_{mid} and w_{mid} , and query the target model. If the model refuses again, the framework performs a second paraphrase iteration to yield $q_{\text{mid}'} \leftarrow \text{getMid}(q_{\text{last}}, q_{\text{mid}})$.

If the target model provides a valid response r_{mid} to either bridging attempt, both the bridging query and its successful response are appended to the chat history $\mathcal{H}$. This incremental bridging step parallels the psychological foot-in-the-door phenomenon, wherein compliance with a minor request facilitates subsequent compliance with more severe transgressions. Following a successful bridge, the next step $k + 1$ repeats the evaluation procedure with the original pair (t_i, w_i) . To prevent infinite loops, the same query q_i is repeated for a maximum of 3 steps before advancing to the subsequent round.

If the model refuses both bridging attempts within the same step, the current round’s queries and responses are entirely excluded from the active chat history $\mathcal{H}$. This measure mitigates history pollution and prevents the model from propagating refusal biases into subsequent rounds. The execution then immediately proceeds to the next round with text query t_{i+1} and watermark w_{i+1} (Algorithm 3).

3.2.4. Putting It All Together

As formalized in Algorithm 1, MFW systematically steers the target MLLM from processing benign or marginally

harmful context to generating explicitly disallowed adversarial outputs through an incremental escalation strategy.

The optimization process initializes at step 1 by introducing the primary escalated query q_1 via its corresponding multimodal input pair (t_1, w_1) . Over the subsequent interaction turns, the framework iteratively traverses the sequence from 1 to n , evaluating the model’s response to each successive (t_i, w_i) pair.

The framework dynamically self-corrects based on the model’s feedback at each step:

- **Refusal Mitigation:** Whenever a safety refusal is detected, the framework triggers the SLIPPERYSLOPEPARAPHRASE mechanism (Algorithm 3) to insert an intermediate bridging prompt, maintaining the current prompt state (t_i, w_i) for the subsequent iteration, provided the threshold of 3 maximum attempts has not been breached.
- **Alignment Enforcement:** If a response is successfully generated but fails the semantic alignment check with respect to the intent of the query, the RE-ALIGN module (Algorithm 2) is executed to generate a revised, more heavily aligned variation of the preceding output.

Through this progressive and iterative optimization loop, the target model’s historical outputs serve as essential stepping stones for downstream exploitation. This design mirrors the psychological foot-in-the-door phenomenon observed in human behavior, wherein compliance with minor concessions significantly increases vulnerability to major ethical transgressions. Ultimately, MFW adapts this behavioral mechanism from the text-only baseline [16] to the complex landscape of multimodal architectures. By seamlessly integrating these modular components, we demonstrate that even robustly aligned MLLMs can be systematically coaxed into generating harmful content, provided that the adversarial escalation is structured with sufficient care and gradual progression.

4. Experiments

4.1. Experimental Setup

Benchmarks. We evaluate all methods on two MAPA-derived safety evaluation settings, denoted as MAPA-HarmBench and MAPA-JailbreakBench. These subsets are sampled from HarmBench [8], a standardized framework for automated red teaming and robust refusal evaluation, and JailbreakBench [3], an open benchmark that defines harmful behaviors, attack artifacts, and standardized scoring functions. Following the MAPA protocol [5], MAPA-HarmBench contains 60 harmful tasks sampled from six semantic categories, while MAPA-JailbreakBench contains 60 harmful tasks sampled from ten categories. This design keeps evaluation tractable while preserving category diversity across the two safety benchmarks.

Models and Baselines. The victim MLLMs are Qwen2.5-

Algorithm 1 Multimodal FITD with Watermark (MFW)

Require: Malicious query q , target MLLM $\mathcal{T}$, attacker model $\mathcal{M}$, max rounds n

Ensure: Jailbroken response or failure state

```
1:  $Q \leftarrow \{q_1, q_2, \dots, q_n\} \leftarrow \text{getEscalatedQueries}(q)$  // Progressive attack strategy
2:  $p_{Is} \leftarrow \text{getSemanticContext}(q)$  // Extract benign visual prompts
3:  $I_{\text{safe}} \leftarrow \text{StableDiffusion}(p_{Is})$ 
4:  $\mathcal{H} \leftarrow \{\}$  // Initialize chat history for  $\mathcal{T}$ 
5:  $i \leftarrow 1$ ,  $\text{step\_count} \leftarrow 0$  // Initialize loop counters
6: while  $i \leq n$  do
7:    $(t_i, w_i) \leftarrow \text{Decompose}(q_i, \mathcal{M})$ 
8:    $I_{\text{mal}} \leftarrow \text{EmbedWatermark}(I_{\text{safe}}, w_i)$ 
9:    $r_i \leftarrow \mathcal{T}(t_i, I_{\text{mal}}, \mathcal{H})$ 
10:  if  $\text{isRejection}(r_i)$  then
11:    if  $\text{step\_count} < 3$  then
12:       $\text{step\_count} \leftarrow \text{step\_count} + 1$ 
13:       $(\mathcal{H}, \text{success}) \leftarrow \text{SSP}(q_i, \mathcal{H}, \mathcal{M}, \mathcal{T}, I_{\text{safe}})$ 
14:      if not  $\text{success}$  then
15:         $\text{step\_count} \leftarrow 0$  // Bridge failed; abandon round
16:       $i \leftarrow i + 1$ 
17:    end if
18:    continue
19:  else
20:     $\text{step\_count} \leftarrow 0$  // Max attempts reached; skip round
21:     $i \leftarrow i + 1$ 
22:    continue
23:  end if
24: end if
25: if not  $\text{isAlign}(r_i, t_i, w_i)$  then
26:    $r_i \leftarrow \text{Re-Align}(t_i, w_i, r_i, \mathcal{H}, \mathcal{T})$ 
27: end if
28:  $\mathcal{H} \leftarrow \mathcal{H} \cup \{(t_i, I_{\text{mal}}), r_i\}$ 
29:  $\text{step\_count} \leftarrow 0$ 
30:  $i \leftarrow i + 1$ 
31: end while
32: return Last response in  $\mathcal{H}$ 
```

VL-3B-Instruct and Qwen3-VL-4B-Instruct. Qwen2.5-VL introduces dynamic-resolution vision encoding and strong document, layout, and visual-agent capabilities [2]; Qwen3-VL further improves long-context multimodal reasoning and visual perception in the Qwen-VL family [1]. We compare our Multimodal FITD with Watermark (MFW) against three representative jailbreak baselines. FITD [16] is a text-only multi-turn jailbreak method that progressively escalates malicious intent through bridge prompts and response realignment. HIMRD [14] is a black-box multimodal jailbreak attack that distributes harmful semantics across text and image

Algorithm 2 Re-Align Mechanism

Require: Current textual query t_i , watermark w_i , misaligned response r_i , chat history $\mathcal{H}$, target MLLM $\mathcal{T}$

Ensure: Aligned response r_{aligned}

```
1:  $p_{\text{align}} \leftarrow \text{GetPredefinedAlignmentPrompt}()$  // Addresses multi-turn inconsistencies
2:  $\mathcal{H}_{\text{temp}} \leftarrow \mathcal{H} \cup \{(t_i, w_i), r_i\}$ 
3:  $r_{\text{aligned}} \leftarrow \mathcal{T}(p_{\text{align}}, \mathcal{H}_{\text{temp}})$  // Nudges model behavior via multi-modality
4: return  $r_{\text{aligned}}$ 
```

Algorithm 3 SlipperySlopeParaphrase (SSP)

Require: Current rejected query q_i , chat history $\mathcal{H}$, attacker model $\mathcal{M}$, target MLLM $\mathcal{T}$, template image I_{safe}

Ensure: Updated chat history $\mathcal{H}$, success flag success

```
1:  $q_{\text{last}} \leftarrow \text{ExtractLastQuery}(\mathcal{H})$ 
2:  $q_{\text{mid}} \leftarrow \text{getMid}(q_{\text{last}}, q_i, \mathcal{M})$  // Maliciousness falls between  $q_{\text{last}}$  and  $q_i$ 
3:  $(t_{\text{mid}}, w_{\text{mid}}) \leftarrow \text{Decompose}(q_{\text{mid}}, \mathcal{M})$ 
4:  $I_{\text{mid}} \leftarrow \text{EmbedWatermark}(I_{\text{safe}}, w_{\text{mid}})$ 
5:  $r_{\text{mid}} \leftarrow \mathcal{T}(t_{\text{mid}}, I_{\text{mid}}, \mathcal{H})$ 
6: if  $\text{isRejection}(r_{\text{mid}})$  then
7:   // First bridge failed; attempt second step shift variation
8:    $q_{\text{mid}'} \leftarrow \text{getMid}(q_{\text{last}}, q_{\text{mid}}, \mathcal{M})$ 
9:    $(t_{\text{mid}'}, w_{\text{mid}'}) \leftarrow \text{Decompose}(q_{\text{mid}'}, \mathcal{M})$ 
10:   $I_{\text{mid}'} \leftarrow \text{EmbedWatermark}(I_{\text{safe}}, w_{\text{mid}'})$ 
11:   $r_{\text{mid}'} \leftarrow \mathcal{T}(t_{\text{mid}'}, I_{\text{mid}'}, \mathcal{H})$ 
12:  if  $\text{isRejection}(r_{\text{mid}'})$  then
13:    return  $(\mathcal{H}, \text{False})$  // Both attempts failed; drop tracking
14:  else
15:     $\mathcal{H} \leftarrow \mathcal{H} \cup \{(t_{\text{mid}'}, I_{\text{mid}'}), r_{\text{mid}'}\}$ 
16:    return  $(\mathcal{H}, \text{True})$ 
17:  end if
18: else
19:    $\mathcal{H} \leftarrow \mathcal{H} \cup \{(t_{\text{mid}}, I_{\text{mid}}), r_{\text{mid}}\}$ 
20:   return  $(\mathcal{H}, \text{True})$ 
21: end if
```

modalities and uses heuristic-induced prompts to encourage reconstruction of the original malicious request. MAPA [5] is a multi-turn adaptive prompting attack for LVLMs that alternates text-vision attack actions within each turn and adjusts the attack trajectory across turns using response feedback.

Evaluation Metrics. We report attack success rate (ASR), the standard metric used in prior jailbreak evaluation [3, 5, 8].

Given N harmful tasks, ASR is defined as

$$\text{ASR} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[J(r_i, q_i) = 1], \quad (1)$$

where q_i is the original harmful task, r_i is the final response produced by the victim MLLM, and $J(\cdot)$ is the judge. In our evaluation, we use HarmBench-Mistral-7B-val-cls, the validation classifier released with HarmBench [8], to determine whether the response is both harmful and relevant to the original task. A higher ASR indicates a stronger jailbreak attack and a lower observed safety robustness of the victim model.

Implementation Details. Our implementation follows the multimodal extension described in Sec. 3. For each harmful task, MFW first generates a FITD-style escalation trajectory and then decomposes every escalated query into a textual component and a short visual watermark. The textual component is sent as the user query, while the watermark is rendered onto a benign image generated from a safety-preserving semantic context. This distributes the attack signal across modalities: neither the text nor the image alone needs to expose the complete harmful instruction, but the victim MLLM can still combine the two modalities through multimodal reasoning. If the victim refuses or drifts away from the intended trajectory, MFW invokes SlipperySlopeParaphrase to insert an intermediate bridge query or uses response re-alignment to restore continuity before advancing to the next escalation level.

For fair comparison with MAPA-derived baselines, we use Qwen/Qwen2.5-7B-Instruct [17] as the attacker model for auxiliary red-teaming operations, including escalation generation, query decomposition, intermediate bridge construction, and re-alignment prompts. The same HarmBench-Mistral-7B-val-cls judge and ASR metric are used for all methods. Unless otherwise specified, victim models are decoded deterministically, while attacker-side generation uses a moderate sampling setting to preserve prompt diversity. Benign background images are synthesized using Stable Diffusion [12]; watermarks are overlaid on the generated images before the paired text-image input is submitted to the victim MLLM. We report only aggregate ASR results and omit concrete harmful prompts or generated harmful content.

4.2. Main Results

Table 1 summarizes the main ASR results. Overall, MFW improves over the original text-only FITD baseline on every model–benchmark pair, raising the average ASR from 45.4% to 51.7%. The gains are most visible on MAPA-HarmBench with Qwen2.5-VL-3B, where MFW improves FITD by 10.0 percentage points, and remain consistent on Qwen3-VL-4B, where MFW improves FITD by 6.6 and 5.0 points on

Table 1. Attack success rate (ASR, %) on MAPA-derived HarmBench and JailbreakBench subsets. Columns report Qwen2.5-VL-3B and Qwen3-VL-4B. Best results are shown in **bold**; second-best non-tied results are underlined.

Method	HarmBench		JailbreakBench		Avg.
	Qwen2.5	Qwen3	Qwen2.5	Qwen3	
FITD [16]	50.0	40.0	51.6	40.0	45.4
HIMRD [14]	38.4	62.3	<u>56.0</u>	85.0	<u>60.4</u>
MAPA [5]	71.0	<u>60.0</u>	58.0	<u>55.0</u>	61.0
MFW (ours)	<u>60.0</u>	46.6	55.0	45.0	51.7

HarmBench and JailbreakBench, respectively. These results support the central hypothesis that FITD-style progressive escalation can be transferred from pure text conversations to multimodal interaction trajectories.

At the same time, MFW is not uniformly dominant over specialized multimodal attacks. MAPA achieves the best average ASR of 61.0%, narrowly outperforming HIMRD by 0.6 percentage points, which reflects the benefit of adaptive within-turn action selection and across-turn trajectory refinement. HIMRD is particularly strong on Qwen3-VL-4B, reaching 62.3% on MAPA-HarmBench and 85.0% on MAPA-JailbreakBench, where it substantially outperforms MAPA. This contrast suggests that different attacks exploit different vulnerability mechanisms: MFW emphasizes gradual compliance and history-dependent self-corruption, HIMRD emphasizes cross-modal risk distribution in a single constructed input, and MAPA emphasizes adaptive search over multiple text-vision attack actions.

The model-wise comparison further reveals that stronger resistance to one attack family does not imply uniform robustness. Qwen3-VL-4B is more resistant than Qwen2.5-VL-3B to MFW, reducing MFW’s average ASR from 57.5% to 45.8%. However, the same model remains highly vulnerable to HIMRD on MAPA-JailbreakBench and to both HIMRD and MAPA on MAPA-HarmBench. This indicates that multimodal safety alignment is both model-dependent and mechanism-dependent: defenses that mitigate gradual FITD-style escalation may still leave exploitable pathways for attacks that more directly distribute or adapt malicious multimodal semantics.

Finally, the dataset-level behavior highlights the importance of evaluating across multiple jailbreak benchmarks. On MAPA-HarmBench, MFW is the second-best method for Qwen2.5-VL-3B, showing that watermark-based multimodalization can provide competitive gains when the target model is susceptible to progressive history construction. On MAPA-JailbreakBench, however, MAPA remains strongest on Qwen2.5-VL-3B, whereas HIMRD dominates on Qwen3-VL-4B. This gap suggests that future multimodal FITD variants should incorporate more adaptive visual-action selection while preserving the gradual escalation structure that makes FITD effective in multi-turn settings.

5. Conclusion

We proposed MFW, extending FITD-style progressive escalation to the multimodal domain by decomposing each escalated query into complementary textual and visual watermark components. Two recovery mechanisms, RE-ALIGN and SLIPPERYSLOPEPARAPHRASE, maintain attack continuity under refusals. Experiments on MAPA-HarmBench and MAPA-JailbreakBench show that MFW consistently improves over the text-only FITD baseline, confirming that gradual escalation transfers effectively to multimodal settings. Our results also reveal that multimodal safety vulnerabilities are both model-dependent and mechanism-dependent, highlighting the need for defenses that jointly address cross-modal reasoning and multi-turn history dynamics.

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